반도체 식각 공정 예측을 위한 시계열 AI 모델

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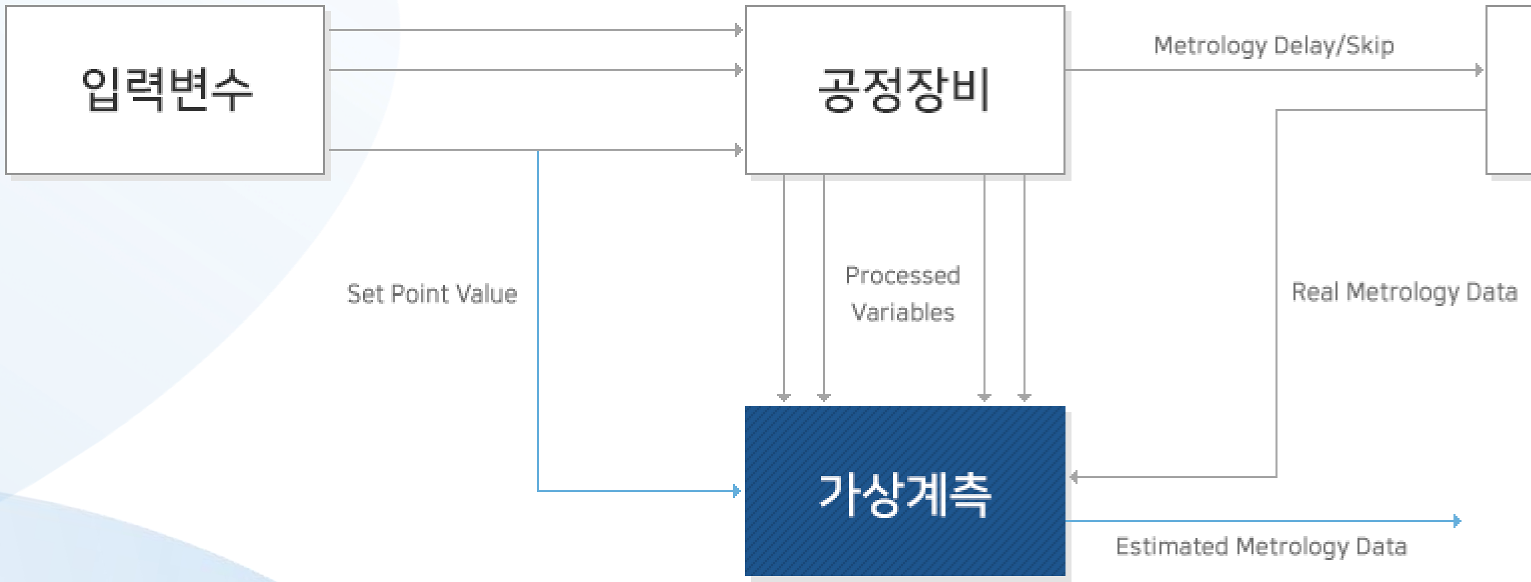
Motivation

- 반도체 제조 공정에서 웨이퍼의 품질 관리는 식각 깊이(etch depth), 산화막 두께(oxide thickness) 등의 물리적 측정에 의존
- 가상 계측(Virtual Metrology, VM)은 이러한 물리적 측정값을 예측하는 기술 → 전수 검사 가능, 공정 제어의 실시간성 향상



VM 선행연구

(1) 기존 OES VM의 한계

- OES는 원래 high-dimensional temporal signal인데 평균, 분산, PCA band 등으로 축약 → temporal dynamics 손실
- PCA를 이용해 주요 wavelength를 선택하고, integration band 및 statistics-based / dynamics-inspired feature를 추출하여 VM 모델을 구축(Chien et al., 2021)

(2) Deep Learning 기반 VM

- 기존 handcrafted feature 대신 representation learning, OES를 2D data 자체로 활용
- DeepVM 연구에서는 convolutional autoencoder를 이용해 OES의 temporal–spectral structure를 자동으로 feature extraction더 높은 예측 성능을 달성((Maggipinto et al., 2019)

Dataset

BOSCH Plasma Etch Multi-Model Dataset

Zenodo | DOI: 10.5281/zenodo.17122442 | CC-BY 4.0

장비

SPTS Omega i2L
DSi Rapier (ICP-RIE)

웨이퍼

200mm Si <100>

공정

BOSCH (SF_6/C_4F_8)
100 cycles

규모

10 lots × 10 wafers ≈ 100

3,648

채널

광학 방출 스펙트럼
185 ~ 884 nm, 25Hz

31

파라미터

공정 센서 데이터
RF Power, 가스, 압력 등

9/89

포인트

웨이퍼 계측값
Etch Depth, Oxide 두께

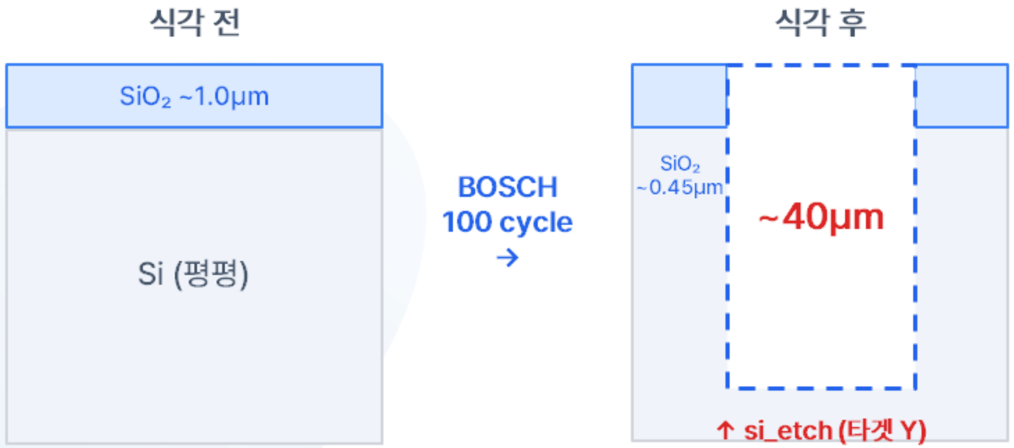
OES + 공정 파라미터(X)로 Etch Depth(Y)를 예측하는 가상 계측(VM) 모델의 학습 데이터

Dataset

타겟 변수 (Y) 및 데이터 파이프라인

무엇을 예측하고, 어떻게 모델에 들어가는가

식각 단면 구조



실제 데이터 (Si_Oxide_etch_9_points.csv)

wafer	loc	preox	postox	si_etch
01	F10	1.003	0.447	40.351
01	D8	0.999	0.363	39.372

데이터 흐름 (파이프라인)

식각 중 (실시간 기록)

공정 파라미터 31개 (5Hz)
OES 3,648채널 (25Hz)

입력 (X)

▼

Feature 추출

통계량 (mean, std, min, max)
OES → PCA 차원 축소

▼

타겟 매칭

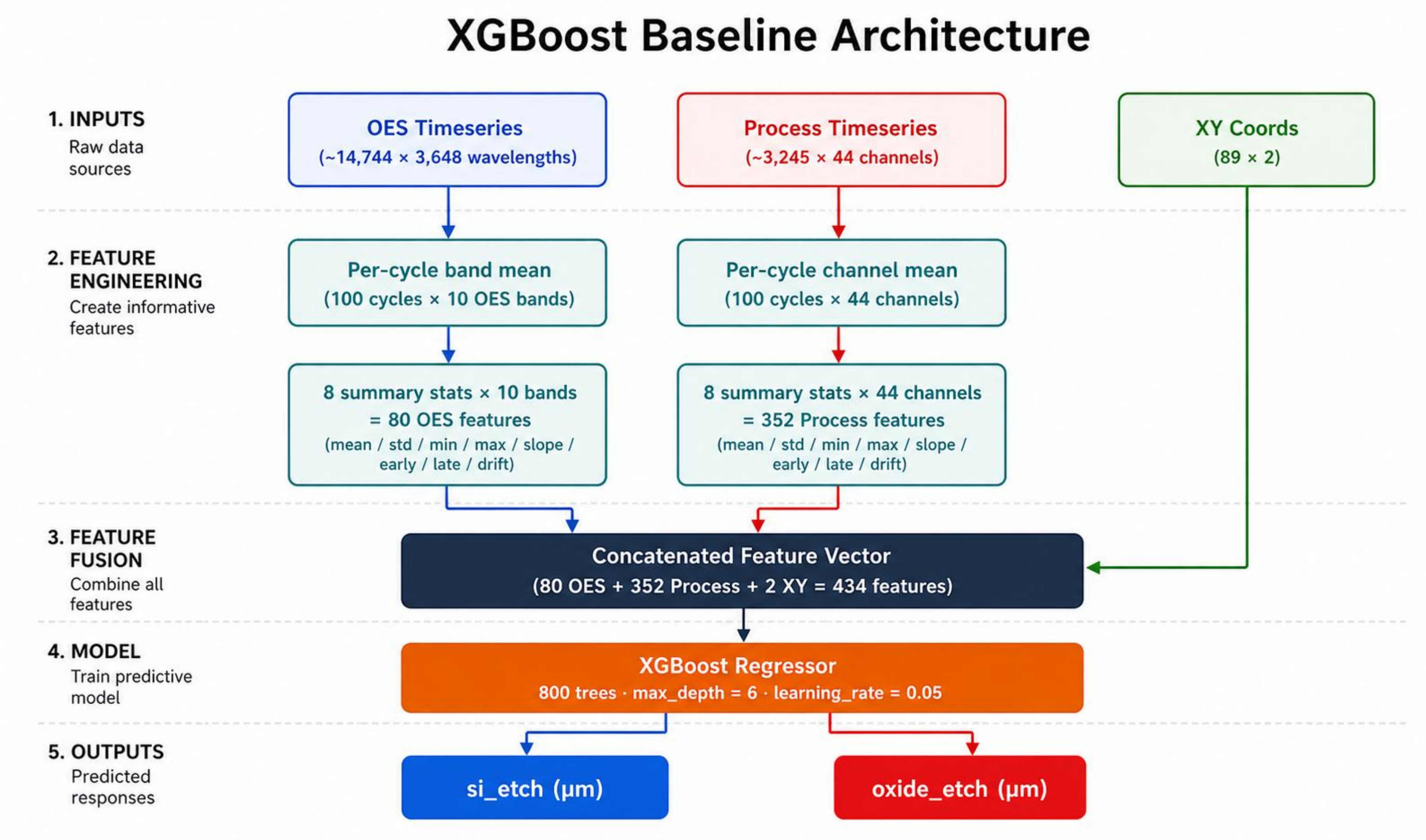
Si Etch Depth (9/89 포인트)
→ 웨이퍼별 평균

타겟 (Y)

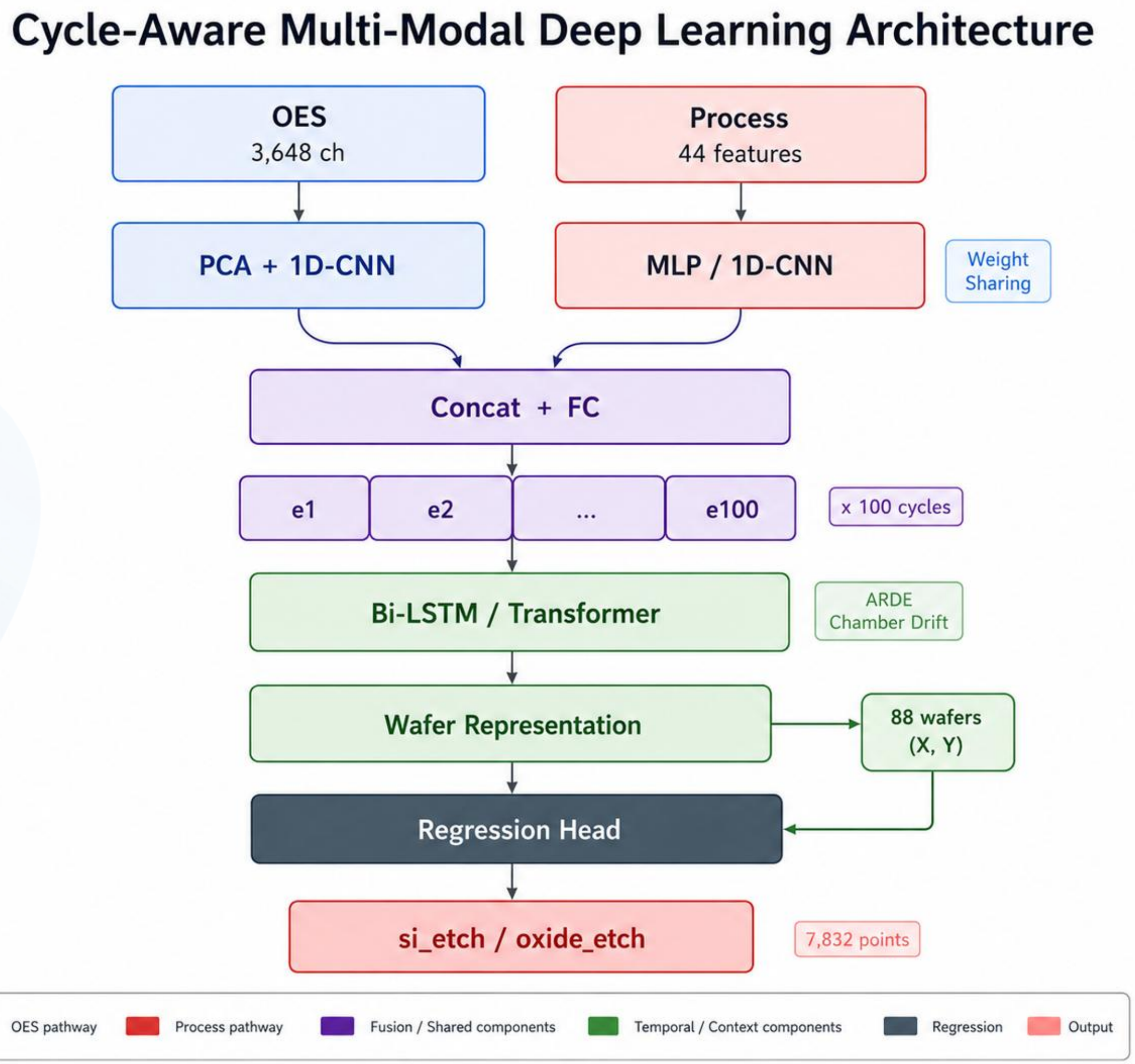
모델 학습

X(Feature) + Y(Etch Depth)
→ ML/DL 가상 계측 모델

Pipeline-XGBoost

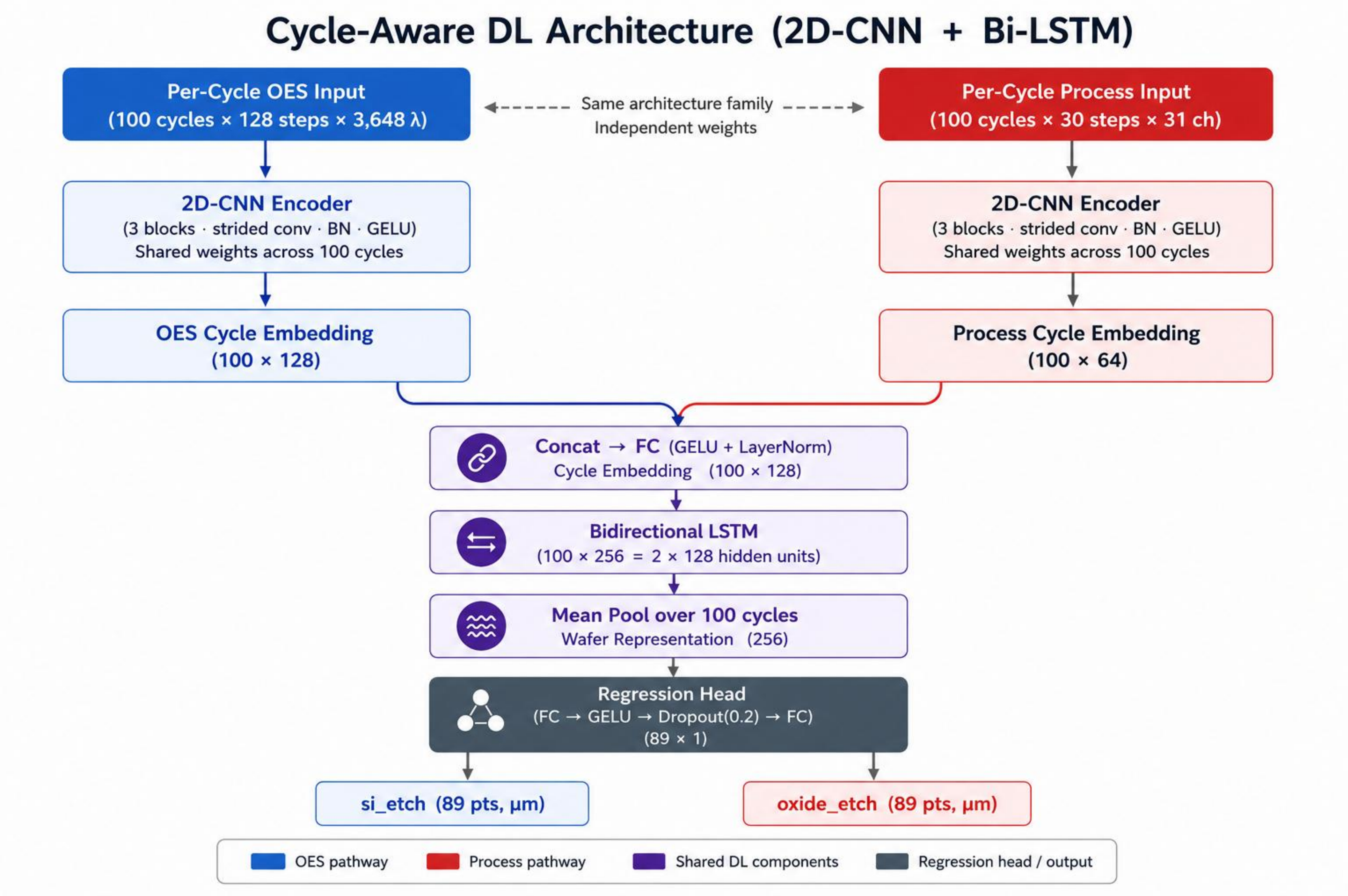


Pipeline-DL



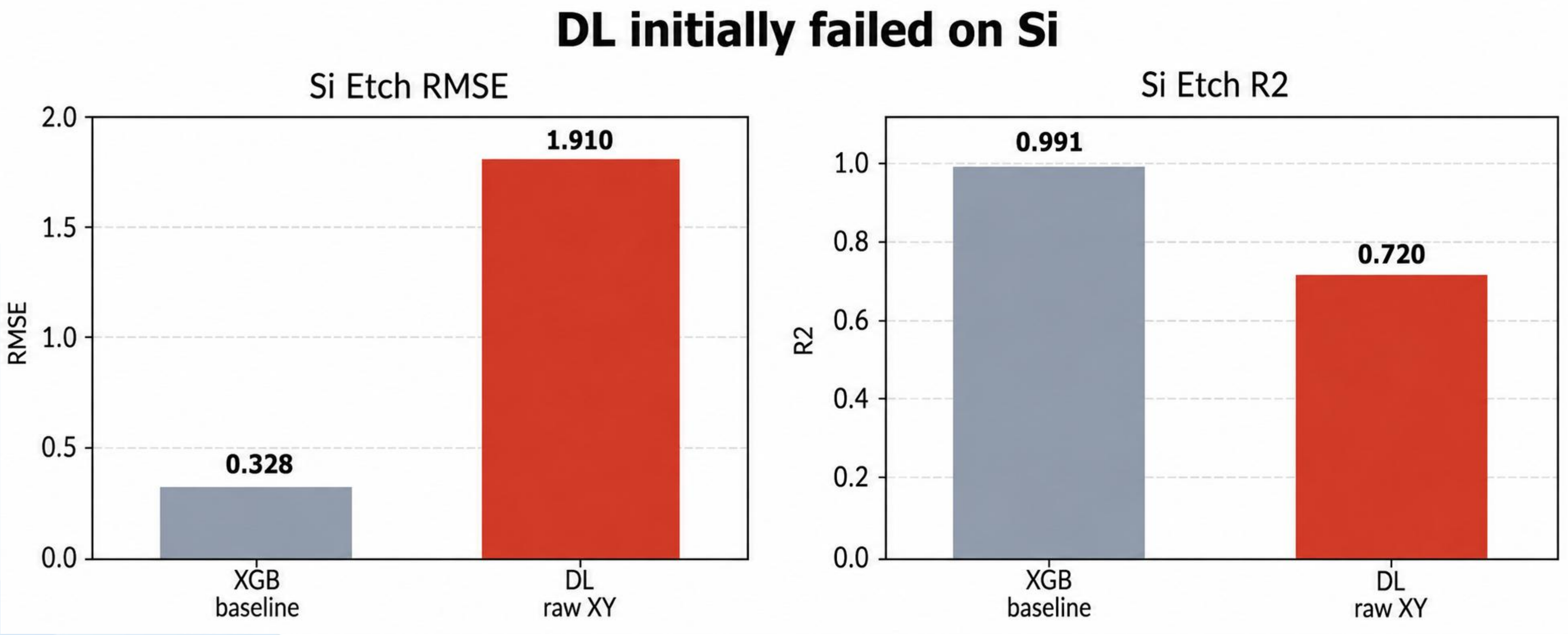
Cycle-Aware Multi-Modal Deep Learning Architecture
OES
3,648 ch
Process
44 features
PCA + 1D-CNN
MLP / 1D-CNN
Weight Sharing
Concat + FC
e1
e2
...
e100
x 100 cycles
Bi-LSTM / Transformer
ARDE Chamber Drift
Wafer Representation
88 wafers (X, Y)
Regression Head
si_etch / oxide_etch
7,832 points
OES pathway
Process pathway
Fusion / Shared components
Temporal / Context components
Regression
Output

Pipeline-DL


Cycle-Aware DL Architecture (2D-CNN + Bi-LSTM)
Per-Cycle OES Input (100 cycles × 128 steps × 3,648 λ)
Same architecture family
Independent weights
Per-Cycle Process Input (100 cycles × 30 steps × 31 ch)
2D-CNN Encoder
(3 blocks · strided conv · BN · GELU)
Shared weights across 100 cycles
2D-CNN Encoder
(3 blocks · strided conv · BN · GELU)
Shared weights across 100 cycles
OES Cycle Embedding (100 × 128)
Process Cycle Embedding (100 × 64)
Concat → FC (GELU + LayerNorm)
Cycle Embedding (100 × 128)
Bidirectional LSTM
(100 × 256 = 2 × 128 hidden units)
Mean Pool over 100 cycles
Wafer Representation (256)
Regression Head
(FC → GELU → Dropout(0.2) → FC)
(89 × 1)
si_etch (89 pts, μm)
oxide_etch (89 pts, μm)
OES pathway
Process pathway
Shared DL components
Regression head / output

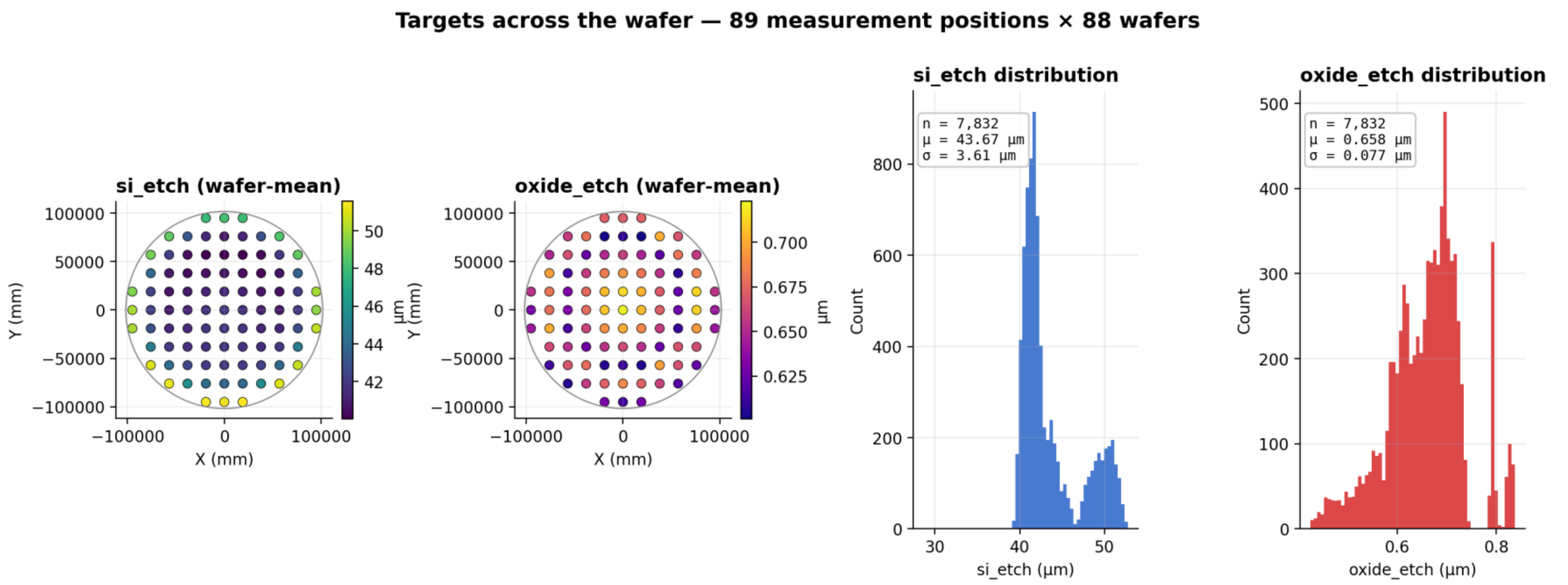
문제1. DL이 Si를 못 맞추는 문제

XGB baseline은 Si를 이미 매우 잘 예측

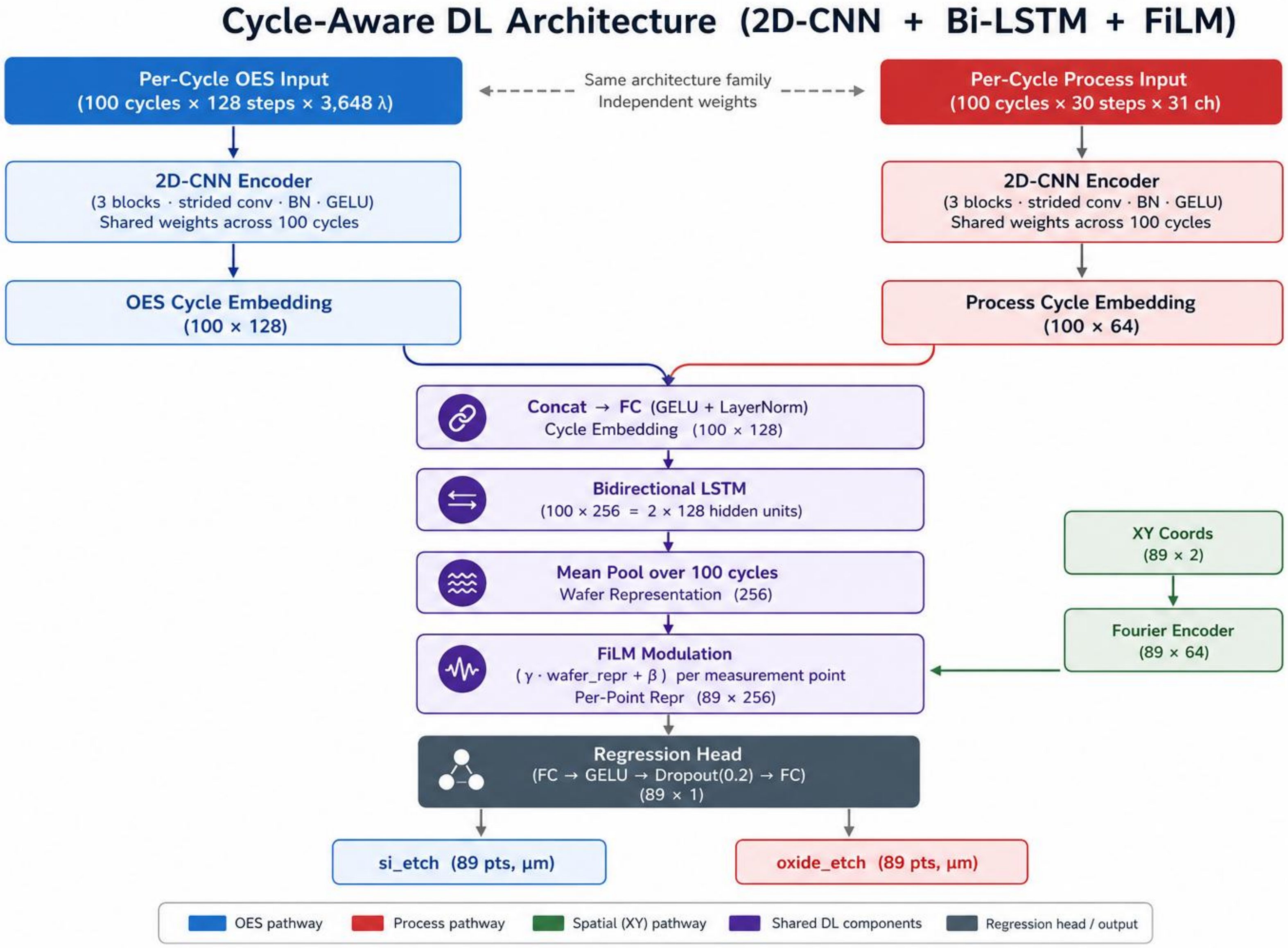


EDA - Si_etch

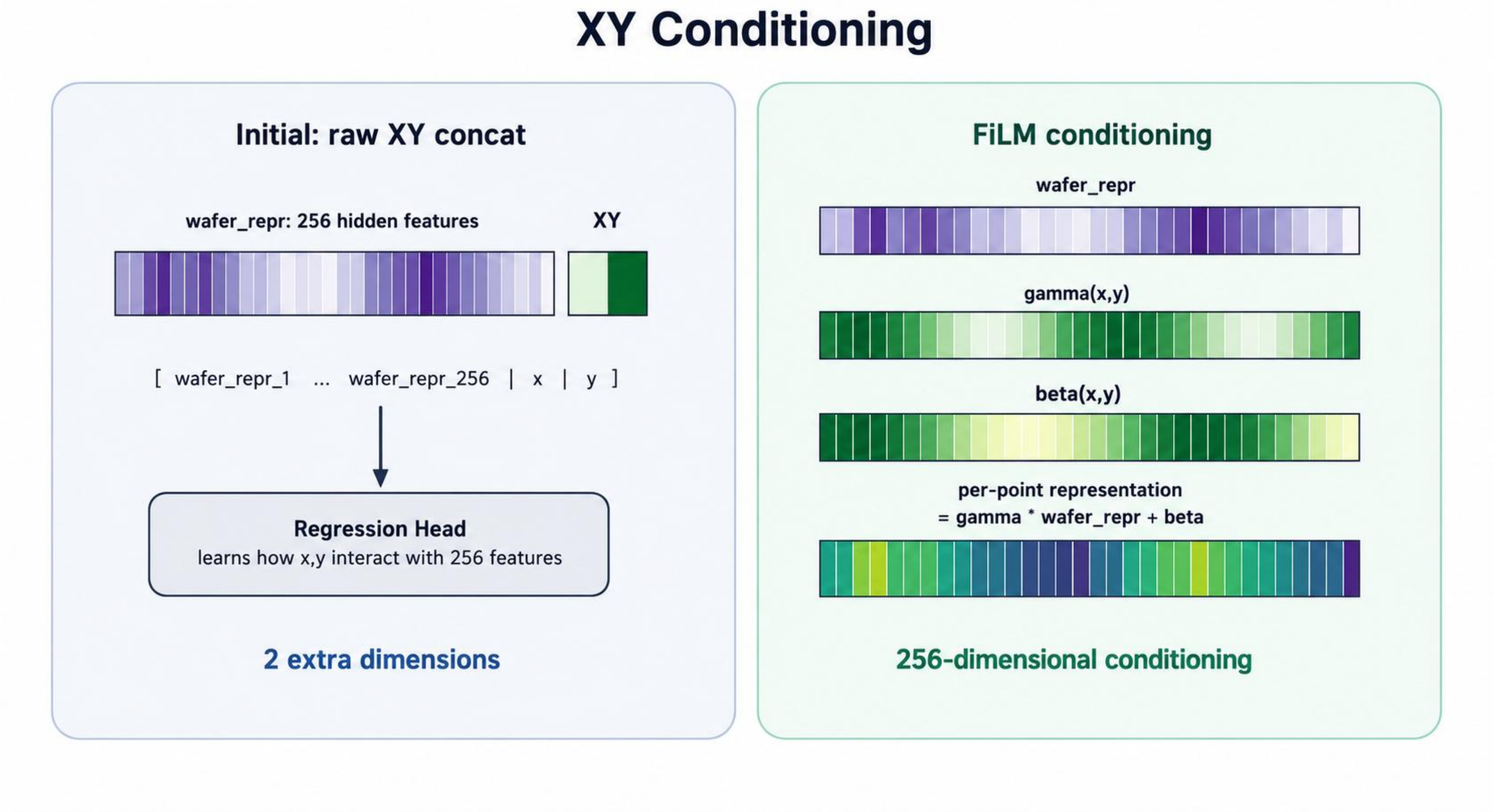
Target	Within-wafer std	Between-wafer std	Spatial baseline R^2
Si_etch	3.61 µm (99%)	0.40 µm (1%)	0.985
Oxide_etch	0.064 µm (68%)	0.044 µm (32%)	0.156



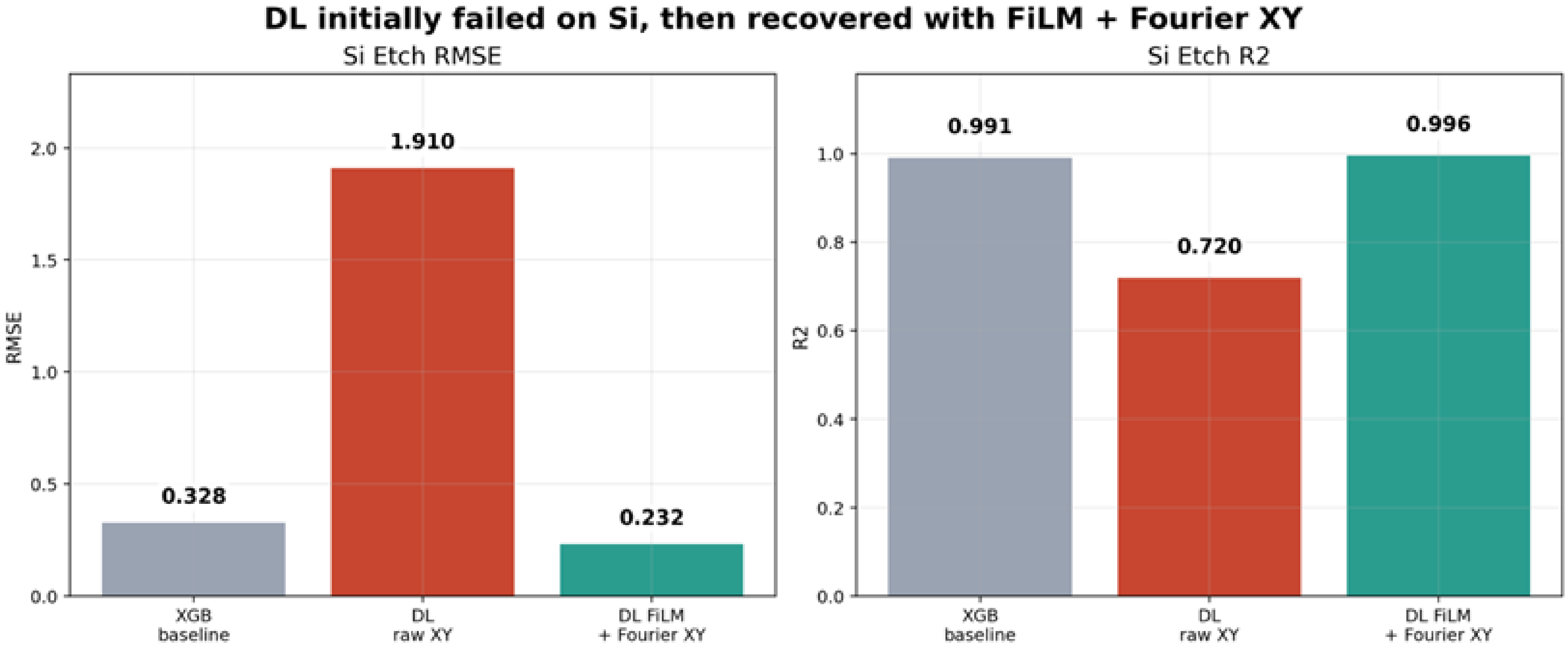
Pipeline-DL


Cycle-Aware DL Architecture (2D-CNN + Bi-LSTM + FiLM)
Per-Cycle OES Input
(100 cycles × 128 steps × 3,648 λ)
Same architecture family
Independent weights
Per-Cycle Process Input
(100 cycles × 30 steps × 31 ch)
2D-CNN Encoder
(3 blocks · strided conv · BN · GELU)
Shared weights across 100 cycles
2D-CNN Encoder
(3 blocks · strided conv · BN · GELU)
Shared weights across 100 cycles
OES Cycle Embedding
(100 × 128)
Process Cycle Embedding
(100 × 64)
Concat → FC (GELU + LayerNorm)
Cycle Embedding (100 × 128)
Bidirectional LSTM
(100 × 256 = 2 × 128 hidden units)
XY Coords
(89 × 2)
Mean Pool over 100 cycles
Wafer Representation (256)
Fourier Encoder
(89 × 64)
FiLM Modulation
(γ · wafer_repr + β) per measurement point
Per-Point Repr (89 × 256)
Regression Head
(FC → GELU → Dropout(0.2) → FC)
(89 × 1)
si_etch (89 pts, µm)
oxide_etch (89 pts, µm)
OES pathway
Process pathway
Spatial (XY) pathway
Shared DL components
Regression head / output

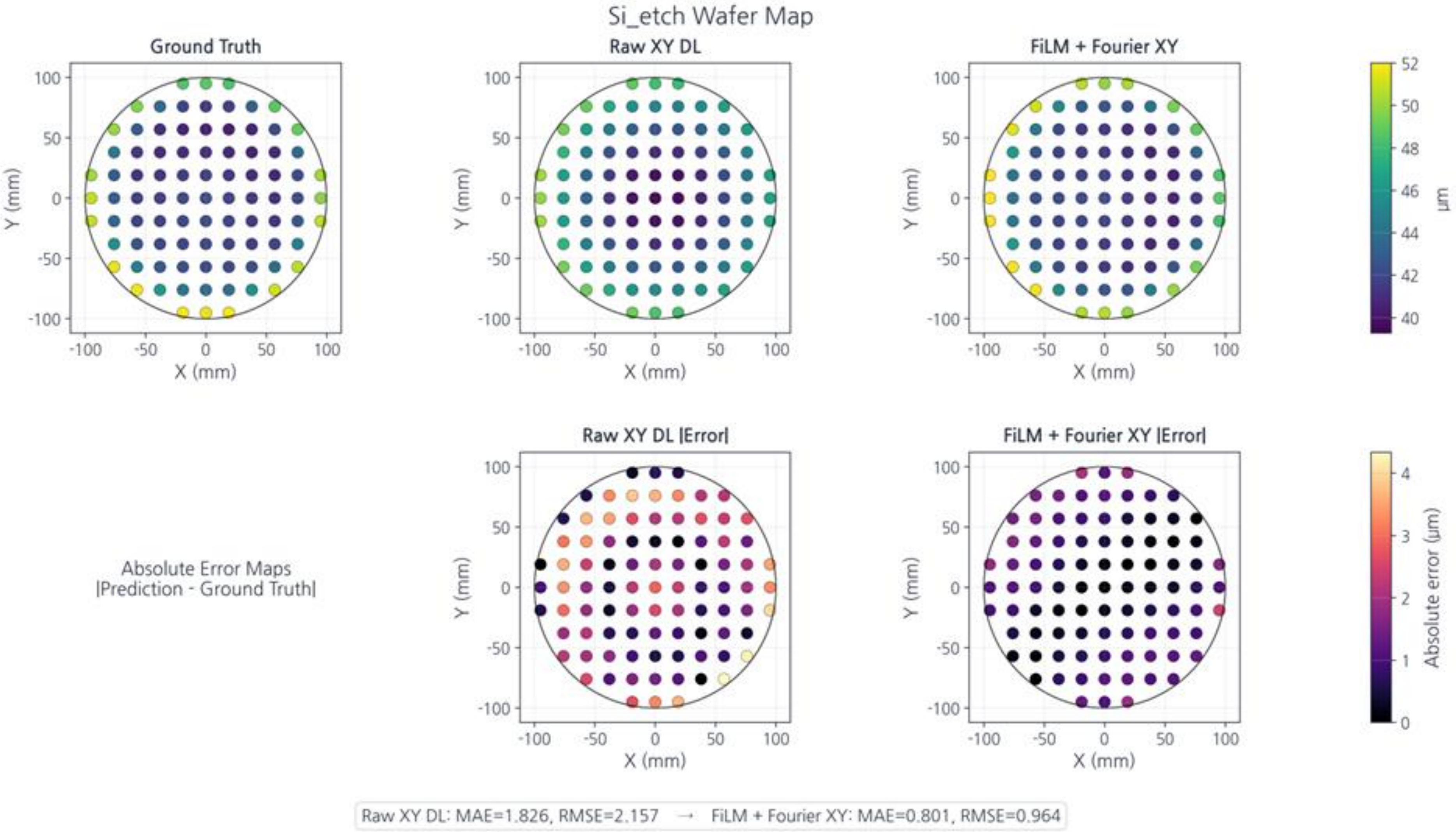
FiLM modulation



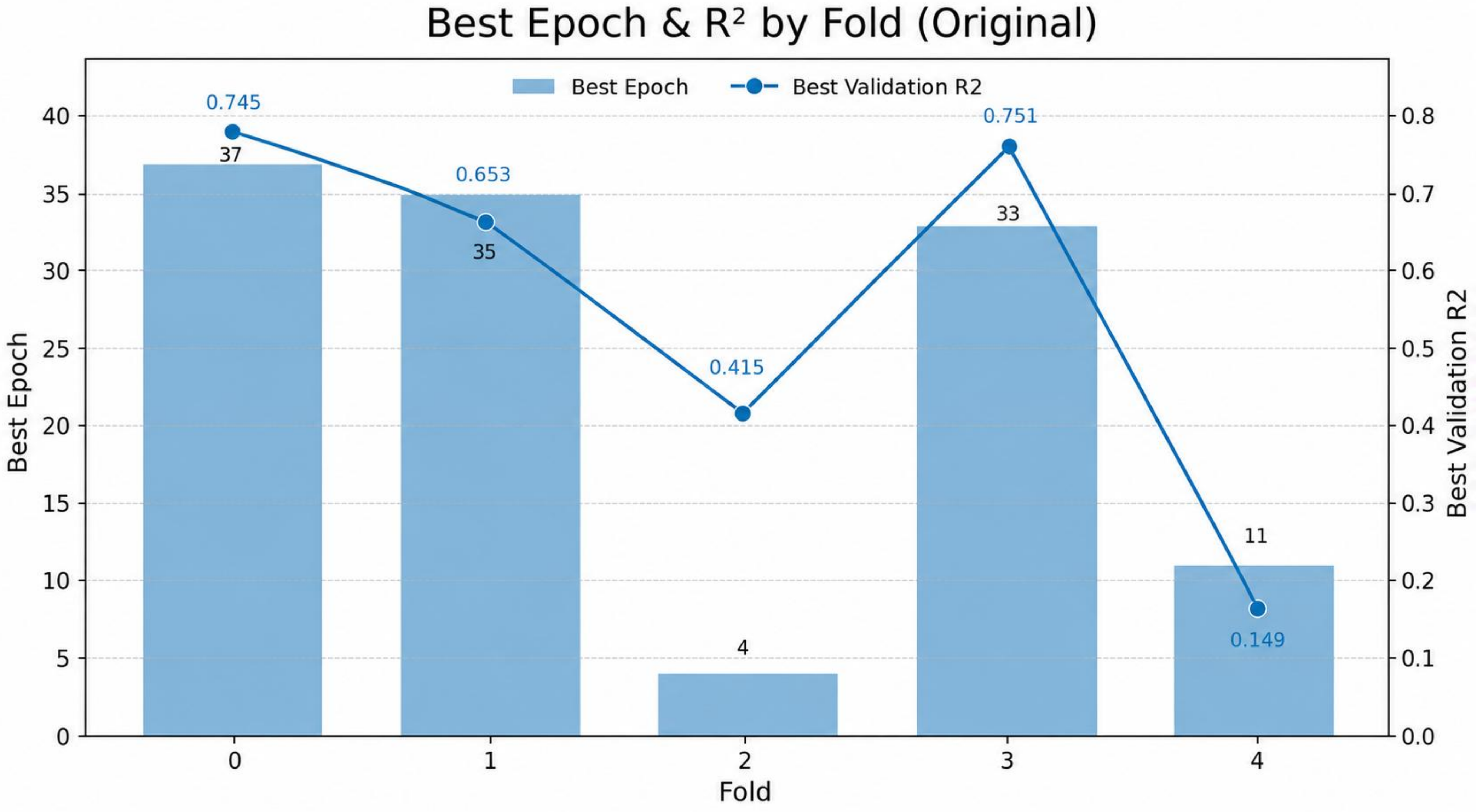
FiLM modulation



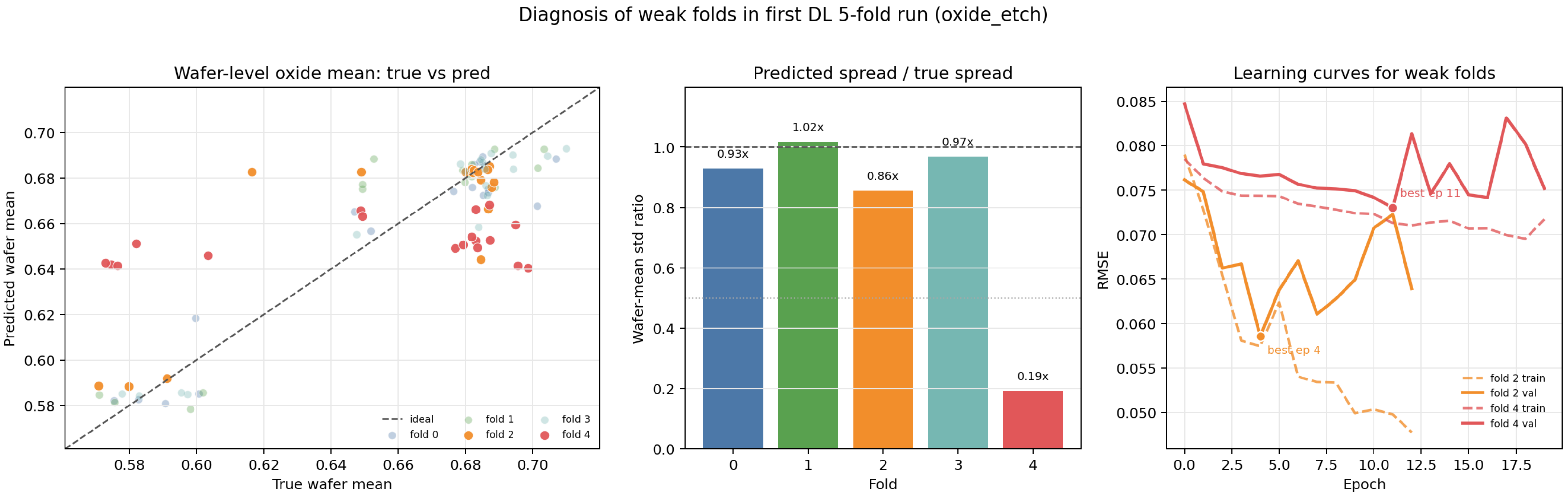
Si_etch wafer-map



문제2. 특정 Fold에서 성능 붕괴

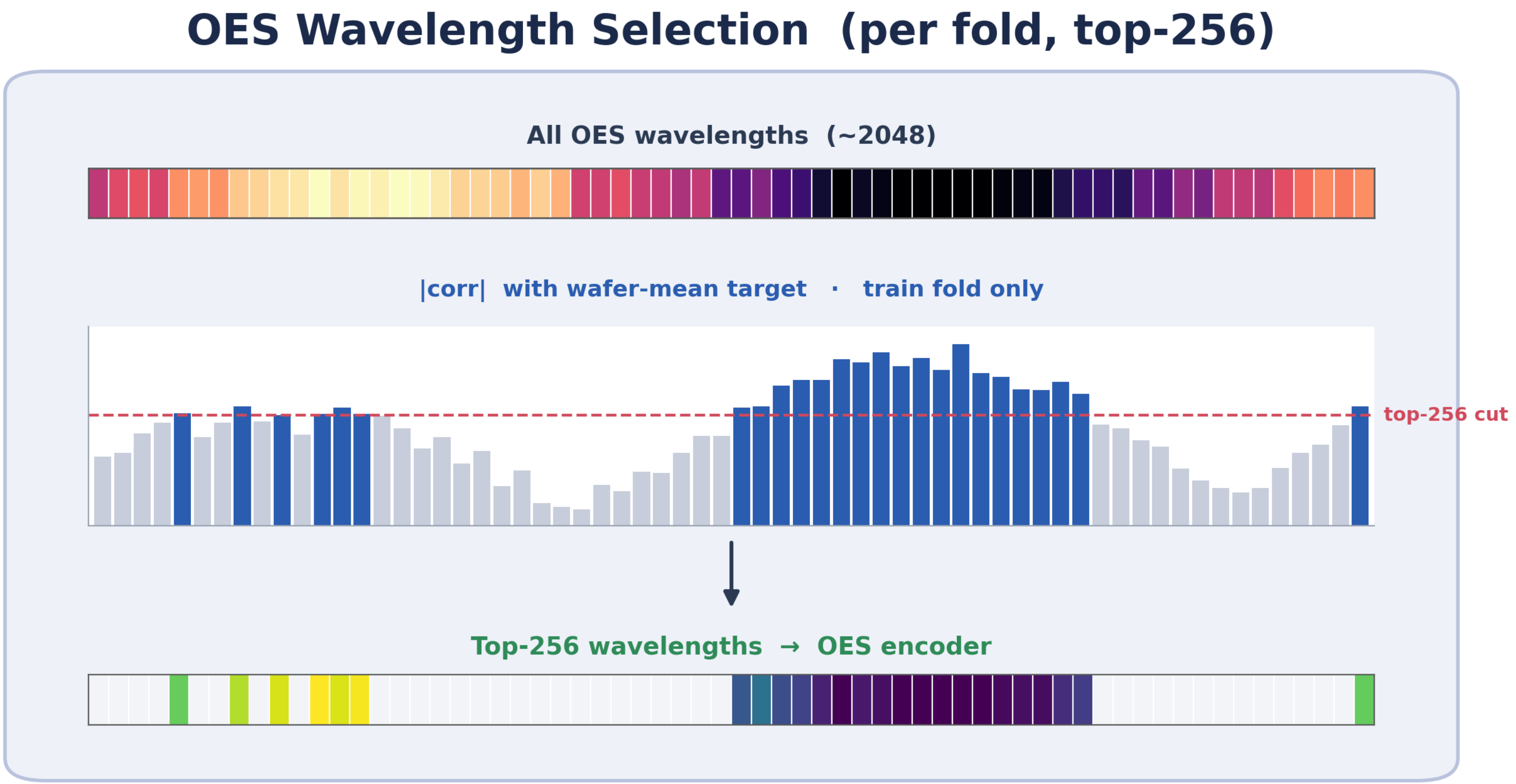


원인 분석



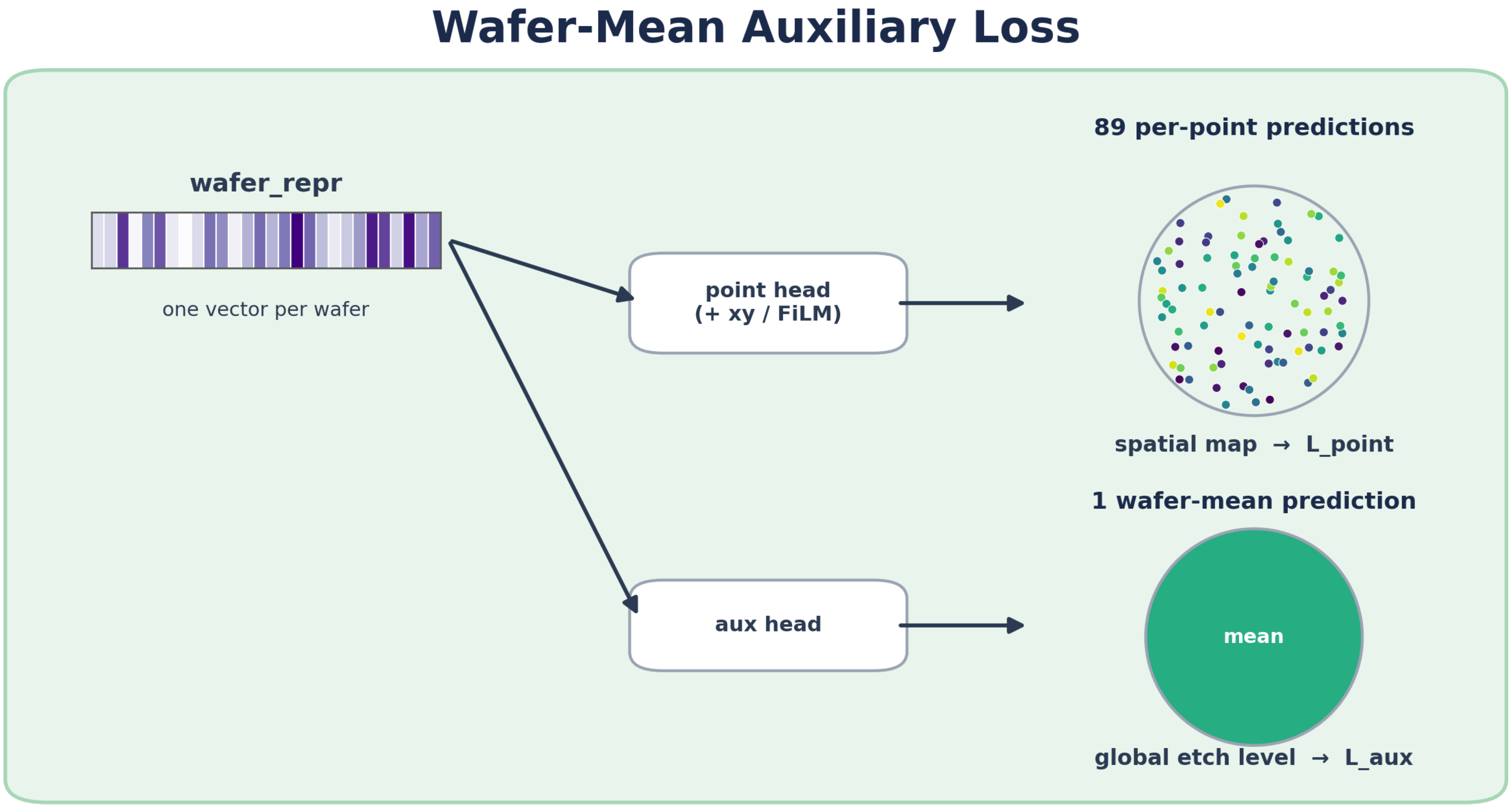
Source: outputs/experiments/2026-05-21_02-09_dl-multimodal-5fold/logs

Improve_1 : OES_top_k



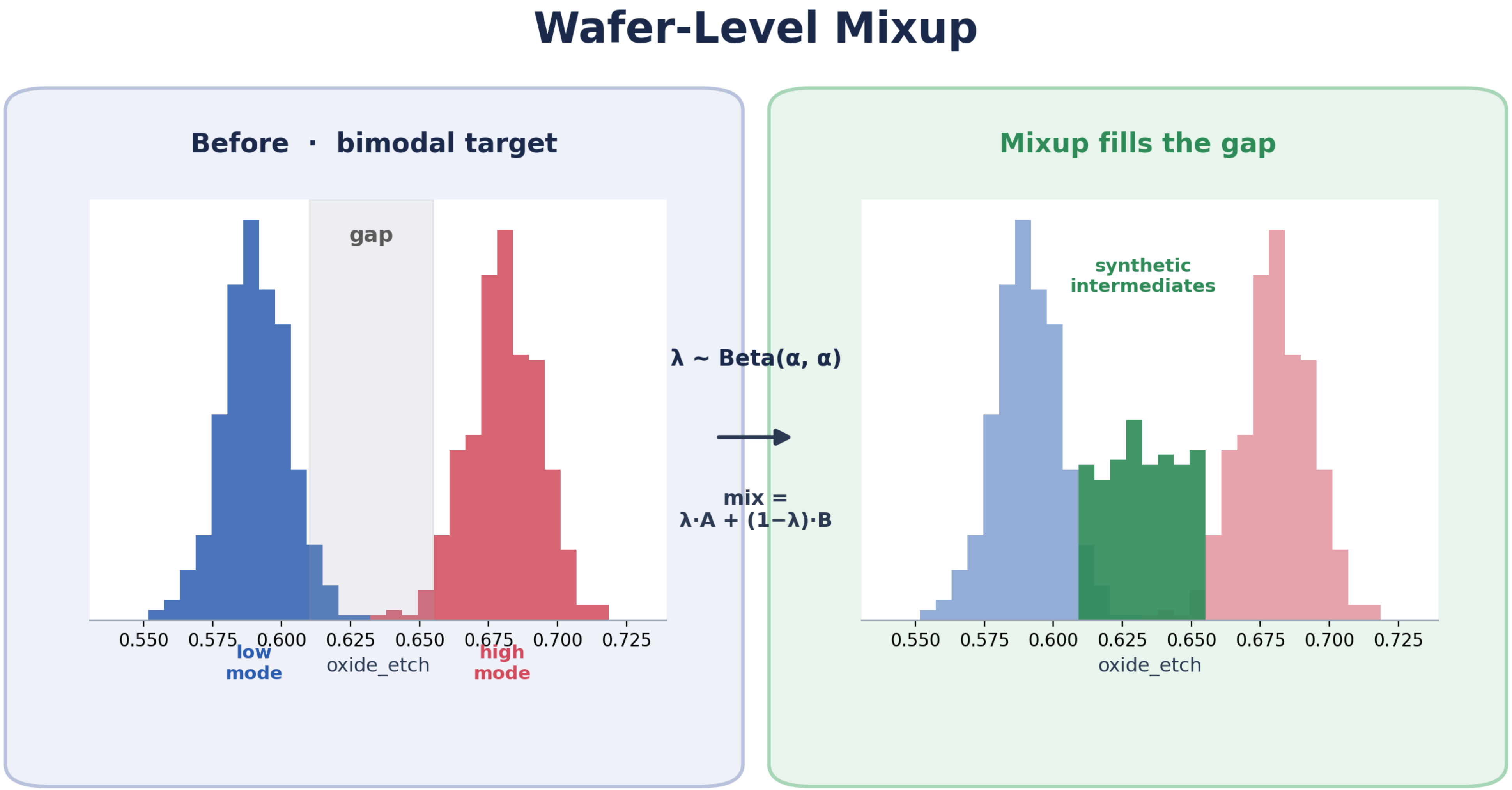
noise wavelengths dropped before the model sees them · no val leakage

Improve_2 : Auxiliary wafer-mean loss



L = L_point + w · L_aux   forces wafer_repr to encode the wafer-level mode

Improve_3 : Mixup

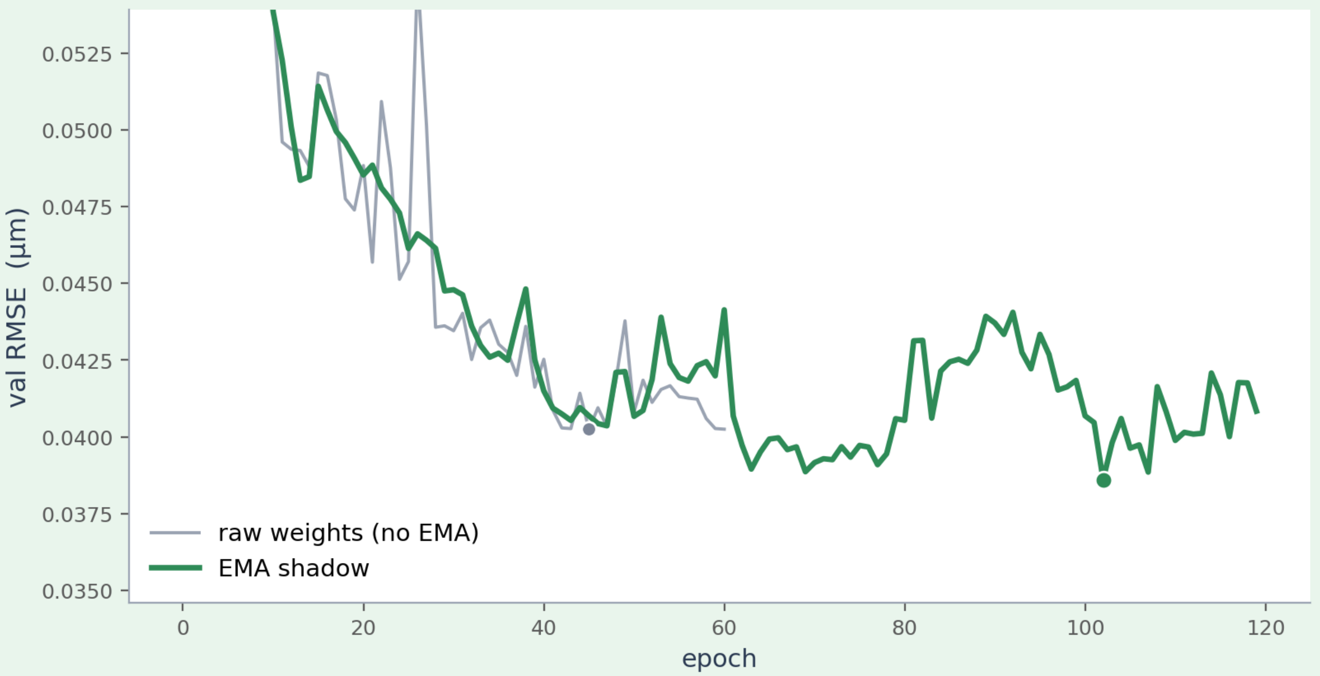


mixes inputs AND target · breaks the mode-collapse basin behind fold 2/4 collapse

Improve_4 : Weight EMA

Weight EMA (Polyak averaging)

fold 0 · oxide_etch



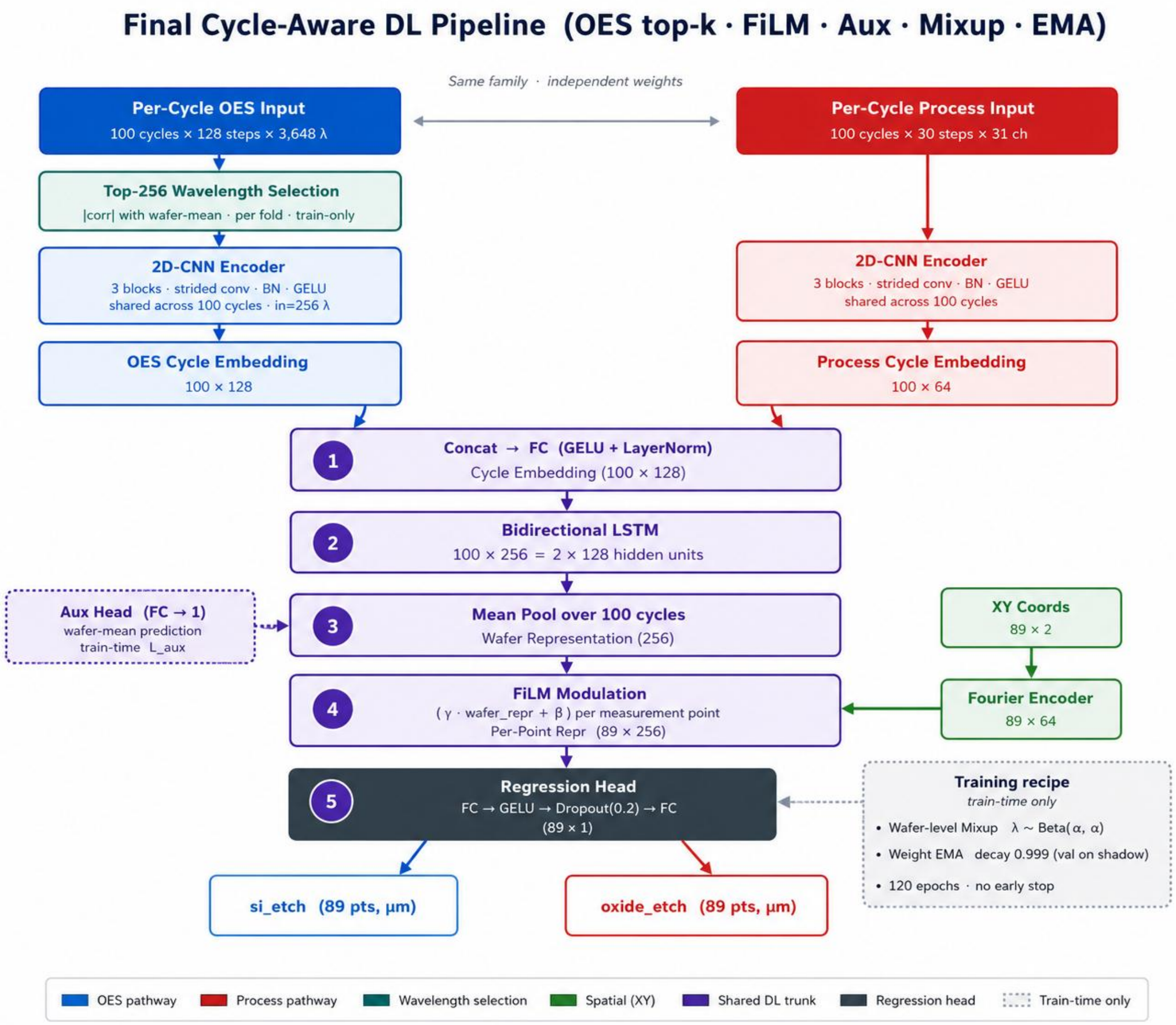
ema ← d·ema + (1−d)·θ

decay d = 0.999

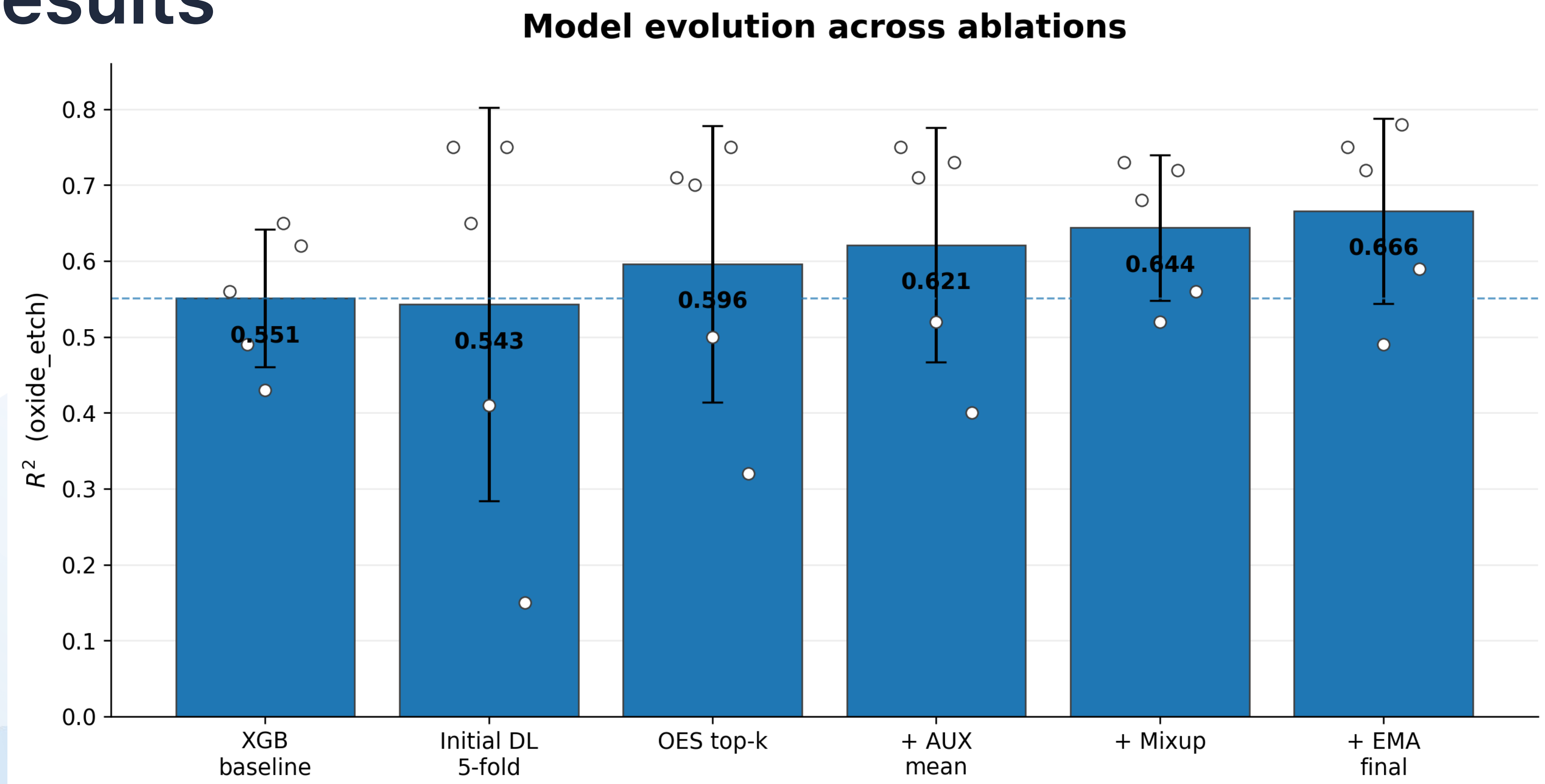
validate / checkpoint on shadow weights

smoother val trajectory · best RMSE 0.0402 → 0.0386 (fold 0)

Final Pipeline

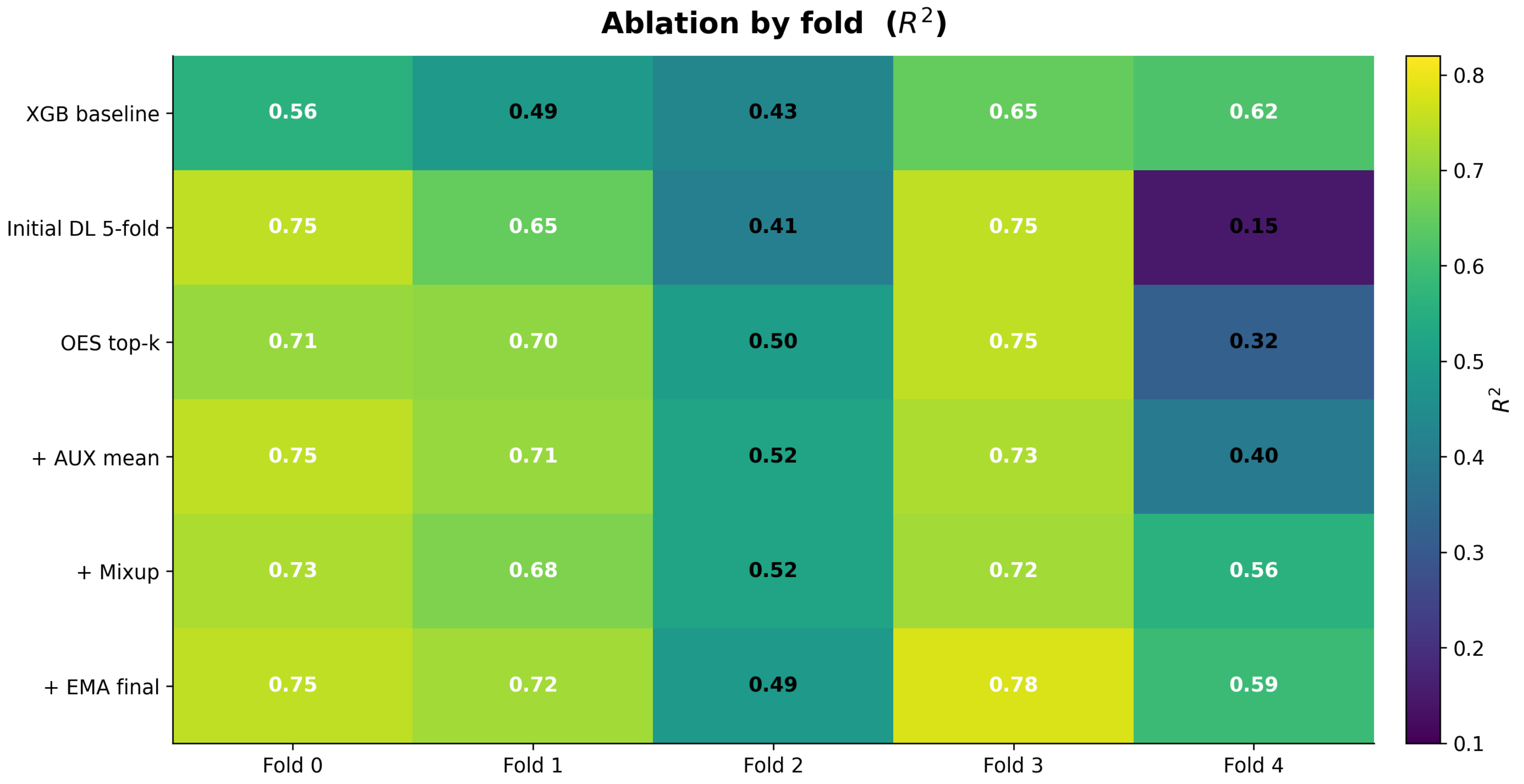


Results



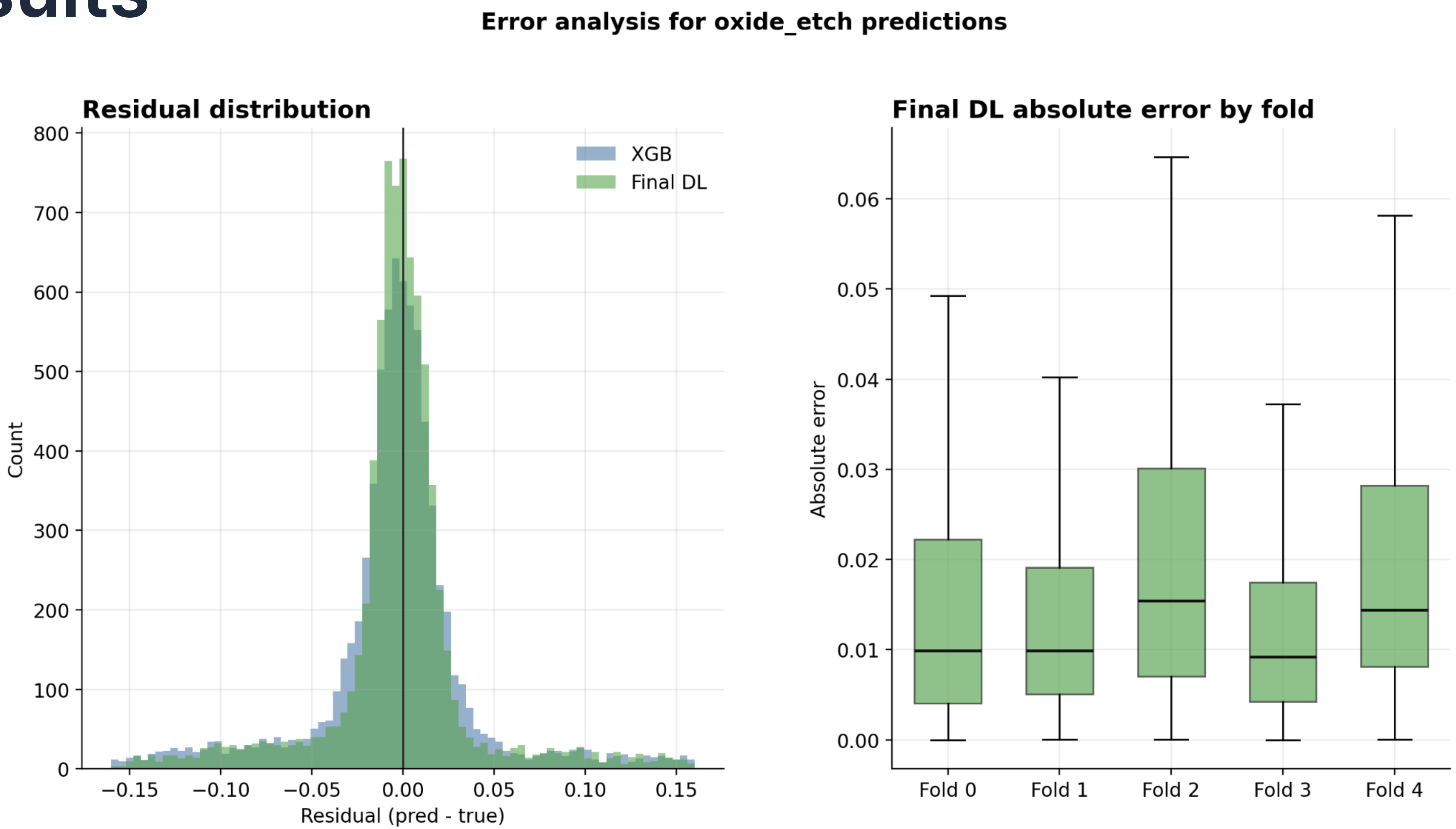
Model evolution across ablations
R^2 (oxide_etch)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.551
0.543
0.596
0.621
0.644
0.666
XGB baseline
Initial DL 5-fold
OES top-k
+ AUX mean
+ Mixup
+ EMA final

Results



Ablation by fold (R^2)
XGB baseline
Initial DL 5-fold
OES top-k
+ AUX mean
+ Mixup
+ EMA final
0.56
0.49
0.43
0.65
0.62
0.75
0.65
0.41
0.75
0.15
0.71
0.70
0.50
0.75
0.32
0.75
0.71
0.52
0.73
0.40
0.73
0.68
0.52
0.72
0.56
0.75
0.72
0.49
0.78
0.59
Fold 0
Fold 1
Fold 2
Fold 3
Fold 4
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
R^2

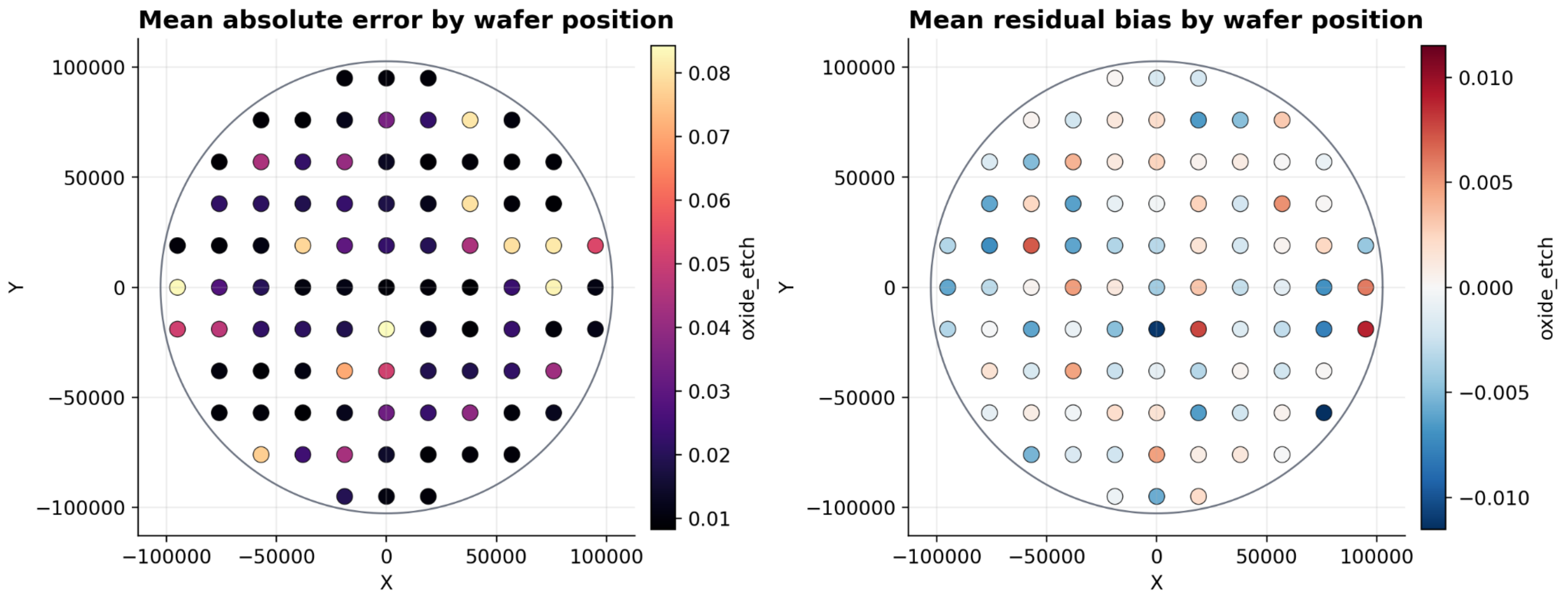
Results



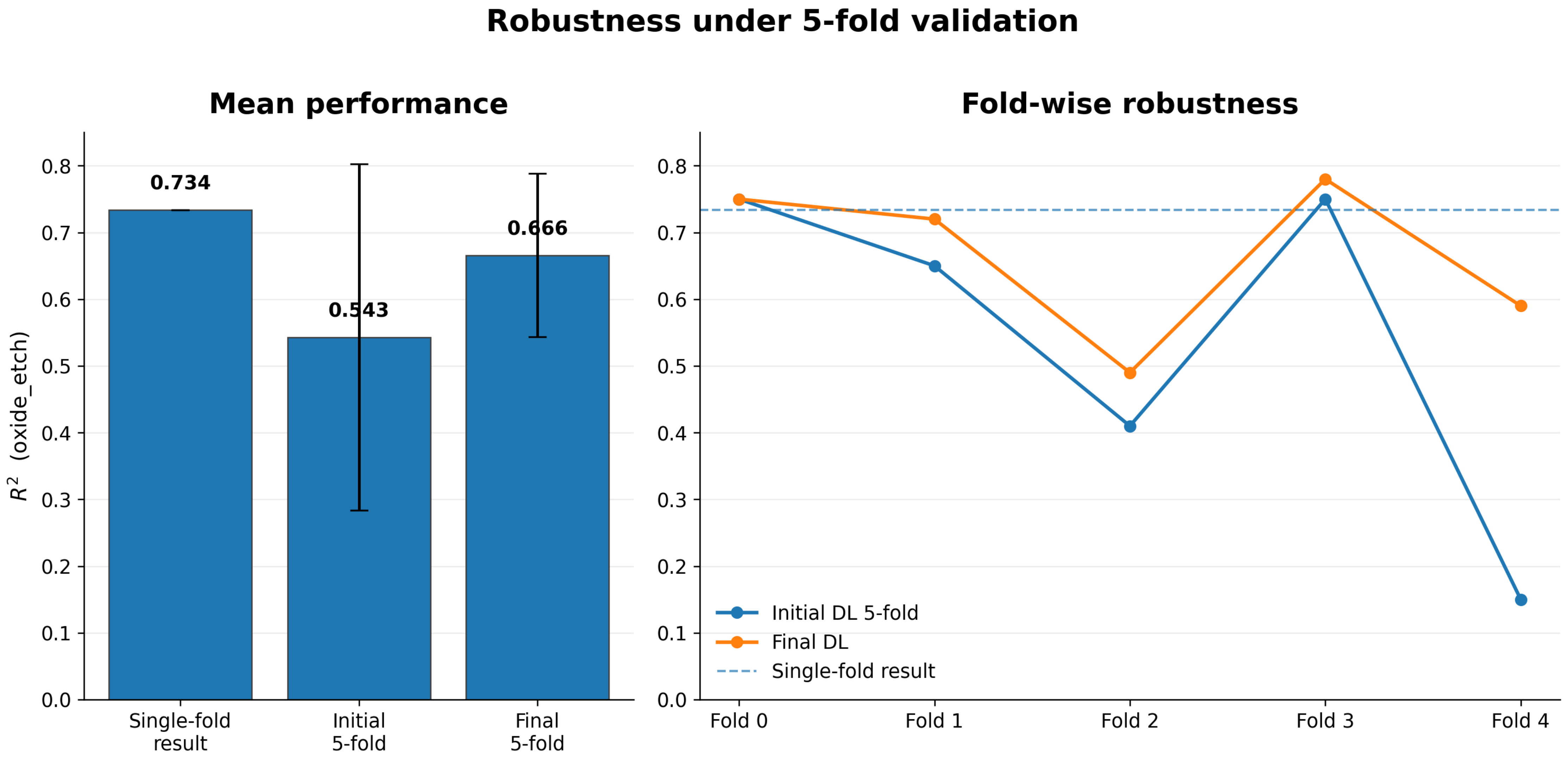
Error analysis for oxide_etch predictions
Residual distribution
XGB
Final DL
Count
Residual (pred - true)
800
700
600
500
400
300
200
100
0
−0.15
−0.10
−0.05
0.00
0.05
0.10
0.15
Final DL absolute error by fold
Absolute error
0.06
0.05
0.04
0.03
0.02
0.01
0.00
Fold 0
Fold 1
Fold 2
Fold 3
Fold 4

Results

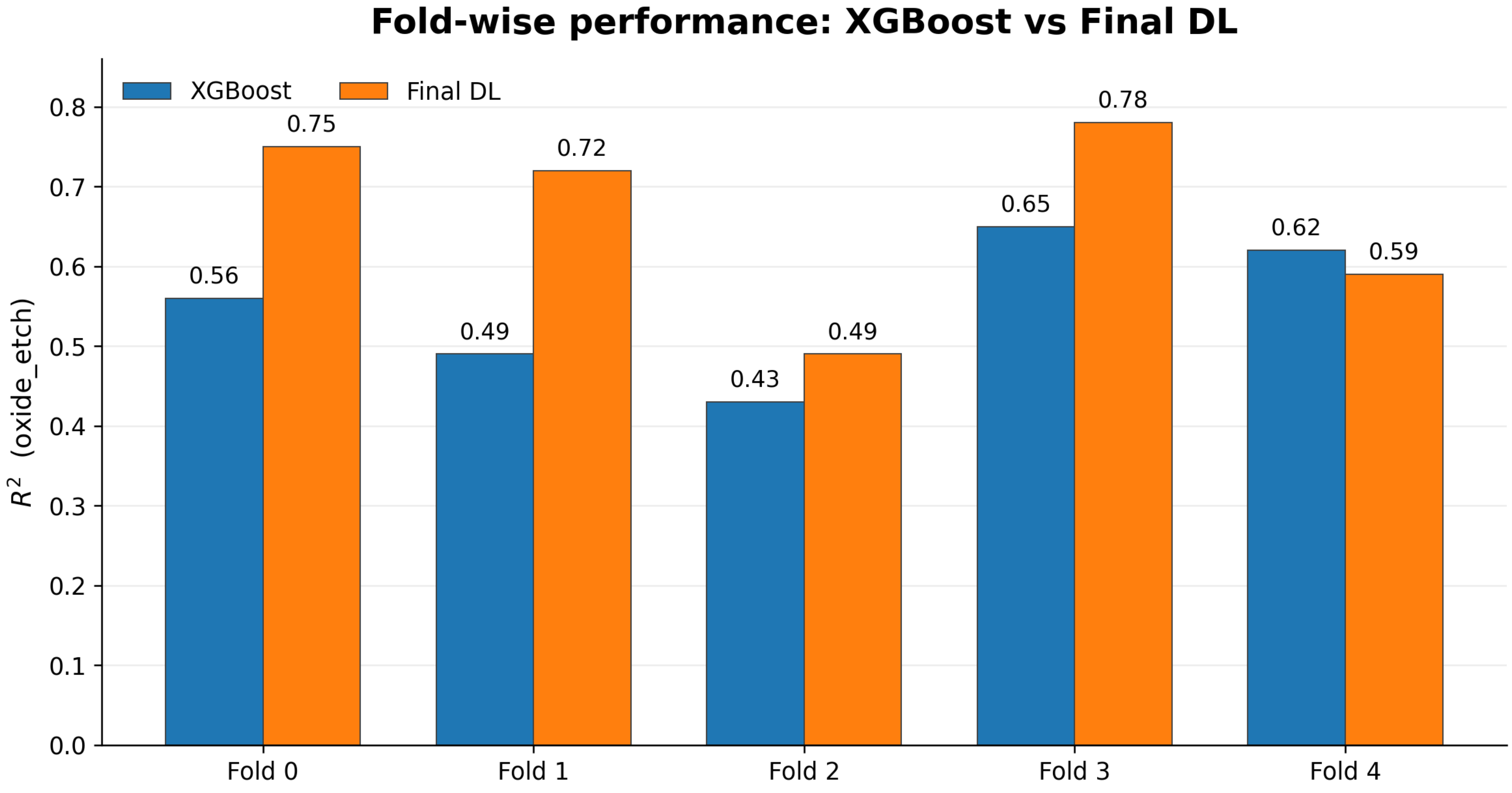
Final DL spatial error map across 89 measurement positions



Results


Robustness under 5-fold validation
Mean performance
Fold-wise robustness
R^2 (oxide_etch)
0.734
0.543
0.666
Single-fold result
Initial 5-fold
Final 5-fold
Initial DL 5-fold
Final DL
Single-fold result
Fold 0
Fold 1
Fold 2
Fold 3
Fold 4
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8

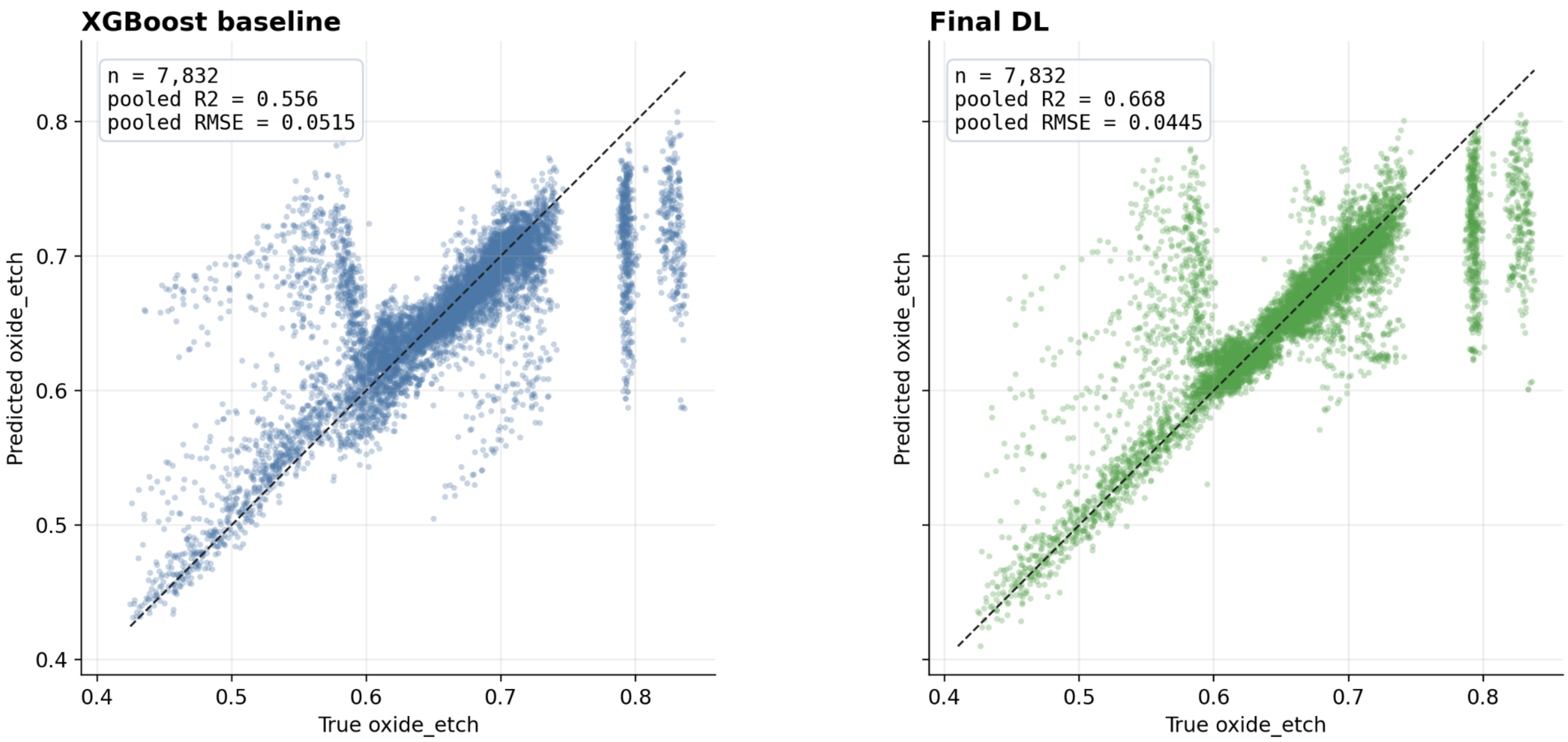
Results



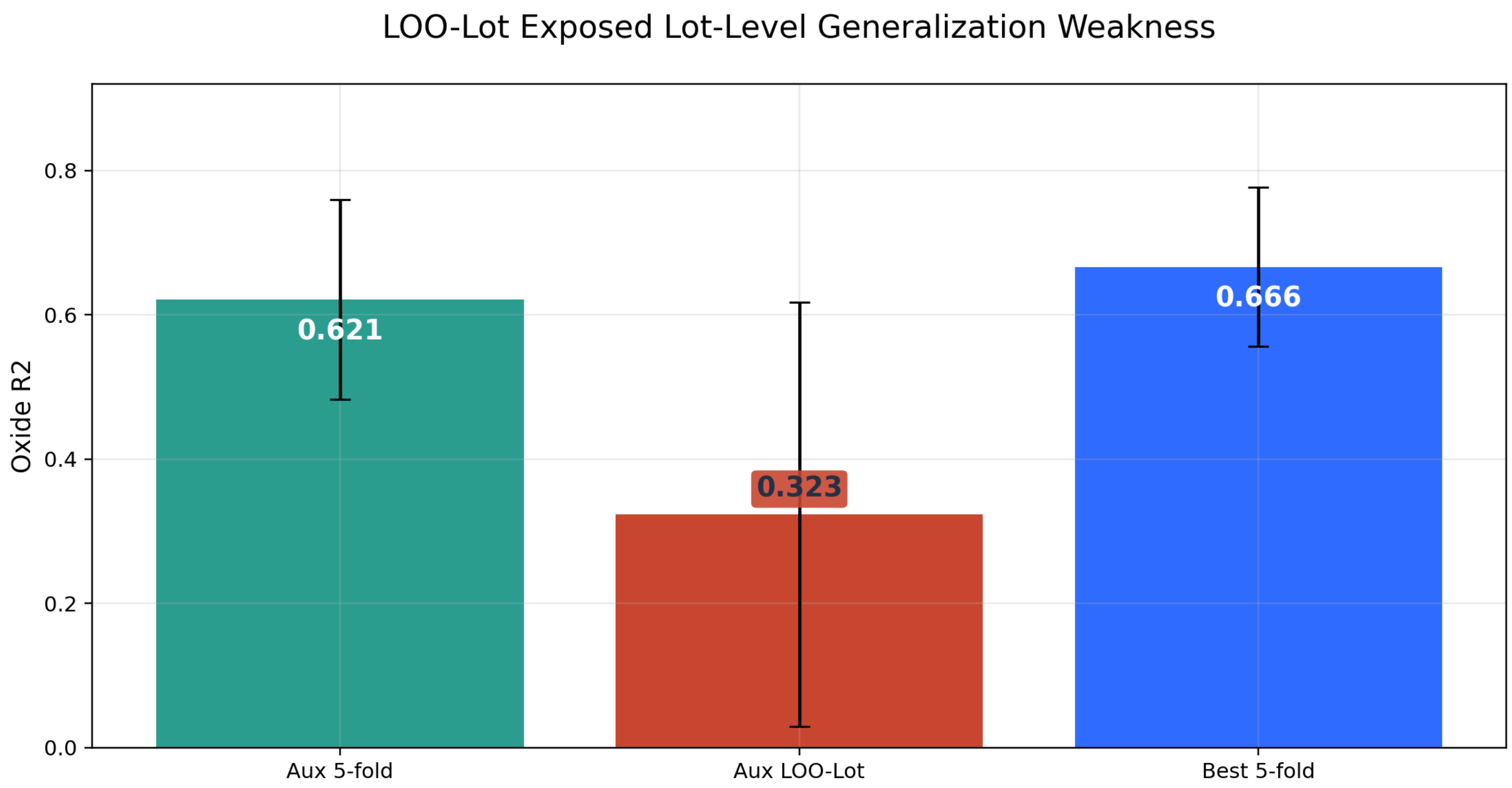
Fold-wise performance: XGBoost vs Final DL
XGBoost
Final DL
R^2 (oxide_etch)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.56
0.75
0.49
0.72
0.43
0.49
0.65
0.78
0.62
0.59
Fold 0
Fold 1
Fold 2
Fold 3
Fold 4

Results

Prediction scatter: all validation points across 5 folds



Limitation


LOO-Lot Exposed Lot-Level Generalization Weakness
Oxide R2
0.0
0.2
0.4
0.6
0.8
0.621
0.323
0.666
Aux 5-fold
Aux LOO-Lot
Best 5-fold

Limitation

1. 데이터 규모와 공정 조건 다양성의 한계
사용 가능한 wafer 수와 공정 조건 범위가 제한되어 있어, 다른 장비나 다른 recipe에 대한 일반화 성능은 아직 검증되지 않음.
실제 양산 환경에서는 chamber drift, maintenance 전후 변화, sensor noise 등이 더 크게 나타날 수 있음.

2. 모델 구조 탐색이 아직 불완전
Multimodal FiLM + Fourier xy + mean pooling 구조가 현재 가장 좋았지만, sequence encoder, pooling 방식, cycle representation에 대한 ablation이 더 필요함.

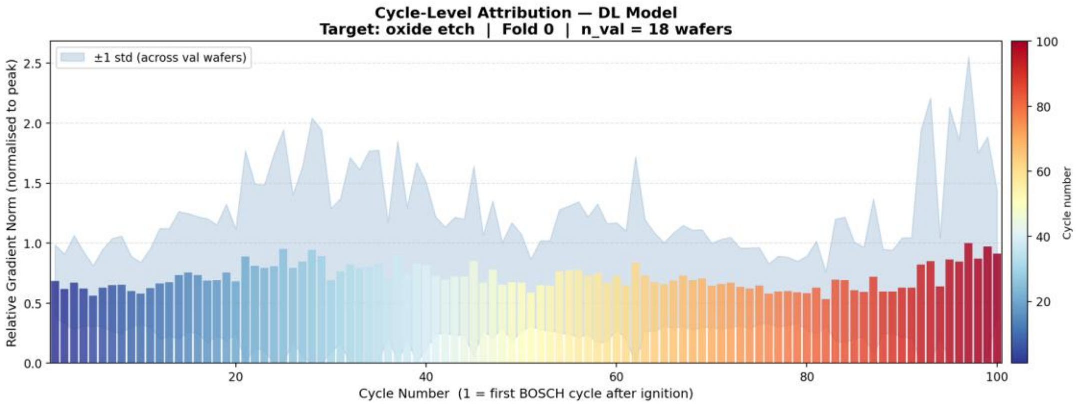
3. VM의 실제 운영 관점 검증 부족
실제 공정 적용을 위해서는 inference speed, wafer별 실시간 예측 가능성, 이상 wafer 탐지와의 연계 등이 추가로 검토되어야 함.

4. 해석 분석이 아직 최종 best 모델 전체로 확장되지 않음
SHAP 및 gradient attribution 분석은 수행되었지만, 최종 best 모델 기준으로 모든 fold에 대해 일관성 있게 정리되지는 않음.
어떤 OES wavelength, cycle 구간, process sensor가 oxide 예측에 중요한지 더 체계적으로 보여줄 필요가 있음.

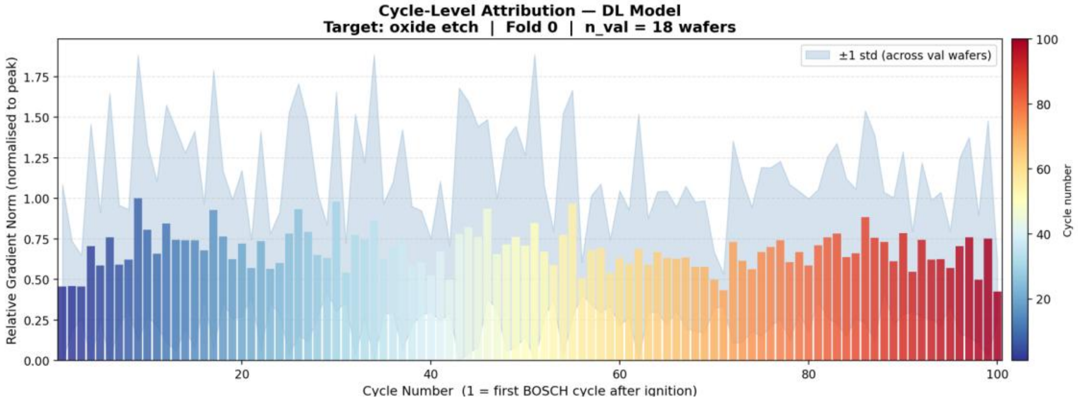
Expected Impact

예측 결과뿐 아니라, 특정 cycle 구간의 이상 신호를 추적하여 공정 drift나 chamber conditioning 문제를 해석할 수 있음

Actual cycle attribution | DL Multimodal | oxide_etch | fold 0 | n_val = 18 wafers



Actual cycle attribution | OES-only | oxide_etch | fold 0 | n_val = 18 wafers



Expected Impact

cycle별 OES/Process signal을 인코딩하고 cycle 간 drift를 학습
->다른 Cycle 공정으로의 확장 가능성

BOSCH DRIE

SF6 **Etch** + C4F8 **Passivation이 반복되는 cycle 공정**

↓

ALD

Gas **공급** + **Purge**가 반복되며 박막을 쌓는 **cycle 공정**

PEALD

Gas **공급** + **Plasma** 처리가 반복되며 박막을 쌓는 **cycle 공정**

ALE

표면 처리 + **얇은 층 제거**가 반복되며 원자층 단위로 깎는 **cycle 공정**

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Thank You

Limitation 해결방안

1. 데이터 규모와 공정 조건 다양성의 한계
• 추가 lot / recipe / chamber 데이터 확보
• maintenance 전후 데이터 포함한 external validation 수행
• LOO-Lot 검증으로 lot-level 일반화 성능 평가

2. 모델 구조 탐색이 아직 불완전
• sequence encoder, pooling 방식, cycle representation 추가 ablation
• top-k OES 선택 기준 및 wavelength window 크기 비교
• aux loss, mixup, EMA 각각의 기여도 분리 검증

3. VM의 실제 운영 관점 검증 부족
• wafer별 inference speed 및 실시간 예측 가능성 평가
• 이상 wafer 탐지 결과와 VM 예측값 연계
• 공정 모니터링 시스템 적용을 위한 threshold / alarm 기준 검토

4. 해석 분석이 아직 최종 best 모델 전체로 확장되지 않음
• 최종 best 모델 기준 5-fold 전체 feature importance 분석
• 주요 OES wavelength, cycle 구간, process sensor 기여도 정리
• fold별 중요 feature 일관성 비교 및 물리적 해석 보강
• ALD · PEALD · ALE 등의 다른 공정에도 반복 cycle 공정으로 확장